

# AI Agent

2026 Fall

소프트웨어융합과

조상구 교수

# 목차

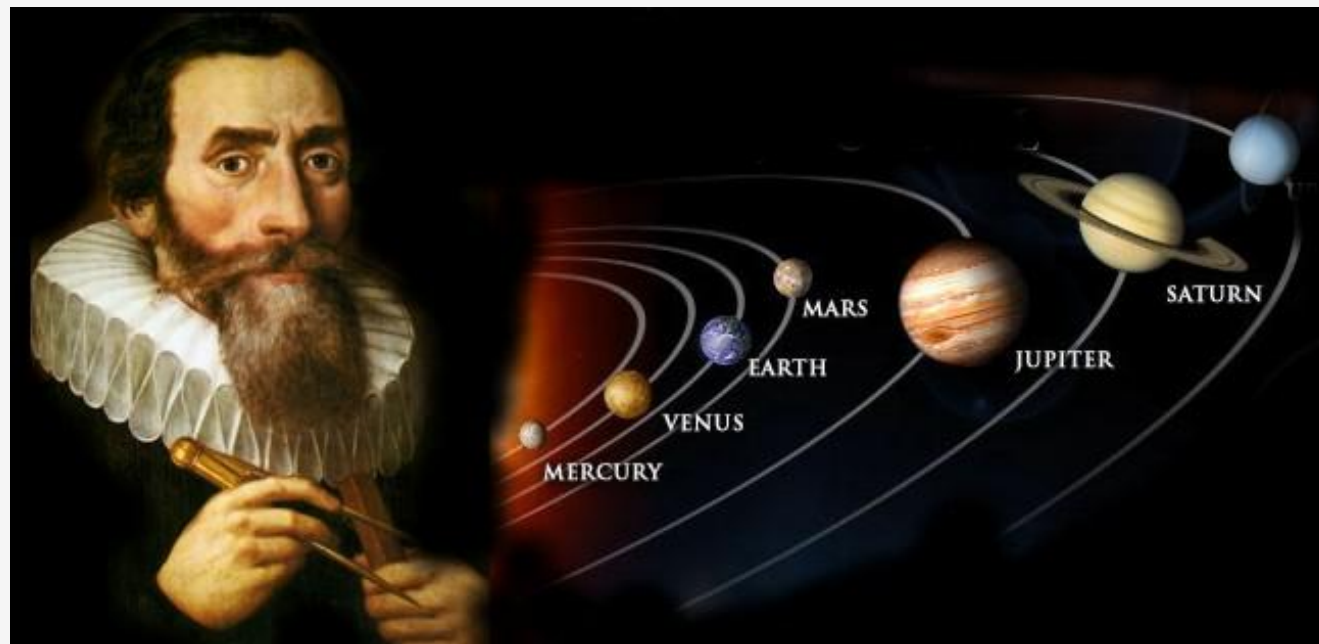
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1. AI는 왜 똑똑하지?
2. AI는 어떻게 학습하지?
3. 생성형 AI
4. LLM agent

# 사람이 규칙을 직접 찾기

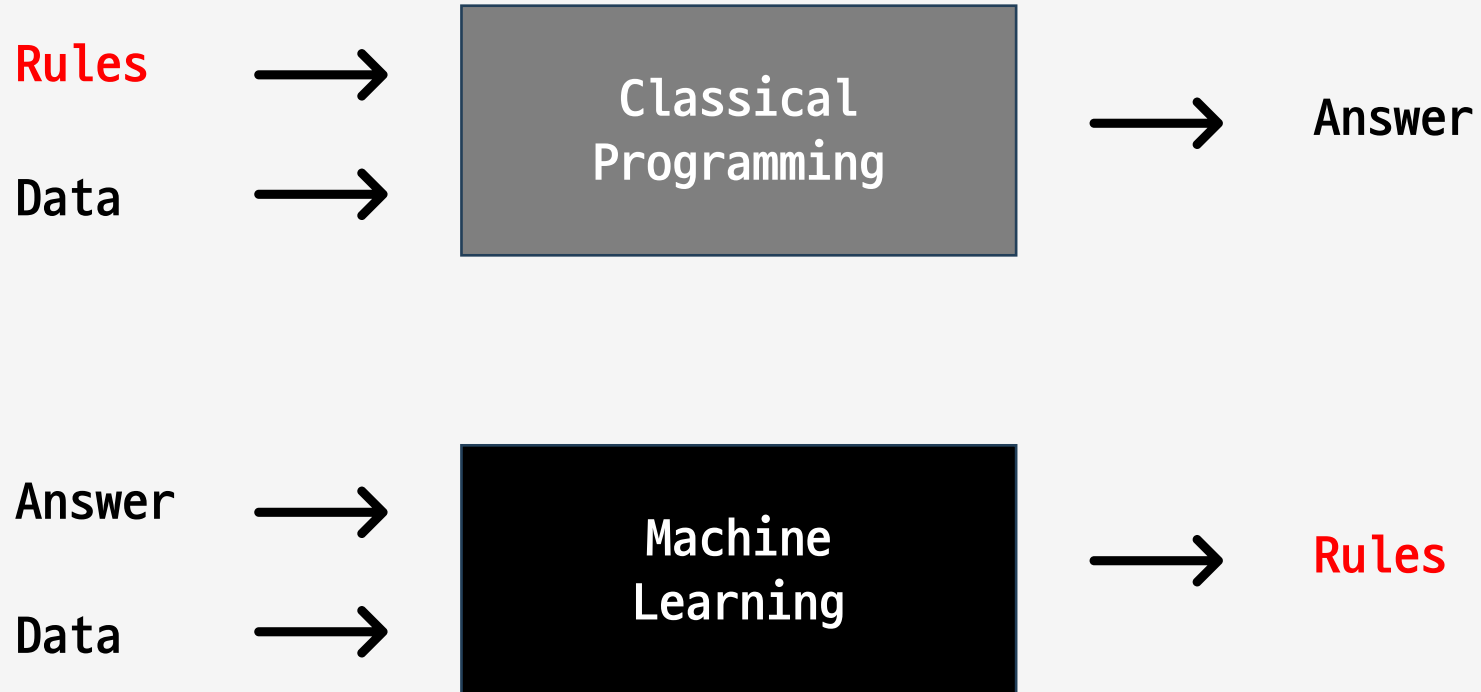
머신러닝이전 시대에는 관측된 데이터에 존재하는 규칙성(Patterns)을 사람이 직접 궁리하고 발견하여 새로운 데이터를 예측 (귀납법)

방대한 데이터들을 직접  
수년간 궁리하고  
계산(Compute)한 끝에,  
행성의 공전궤도는 원이  
아닌 '타원'이라는  
사실(규칙성, 패턴)을 발견

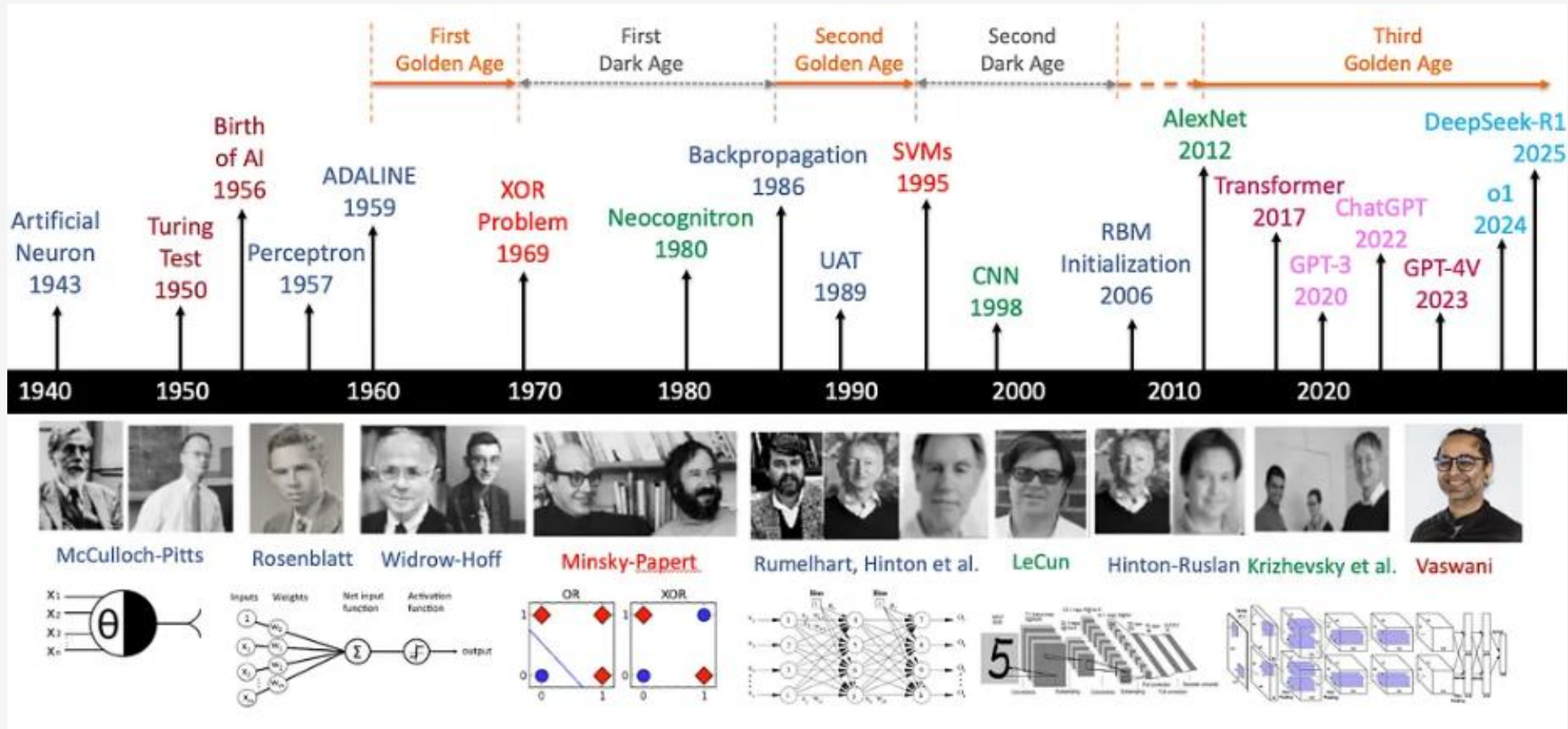


# 머신러닝

과거 데이터에 존재하는 규칙성(Patterns)을 컴퓨터가 발견하게 하여 새로운 데이터를 예측하는 것이  
머신러닝(Machine Learning) → New Paradigm



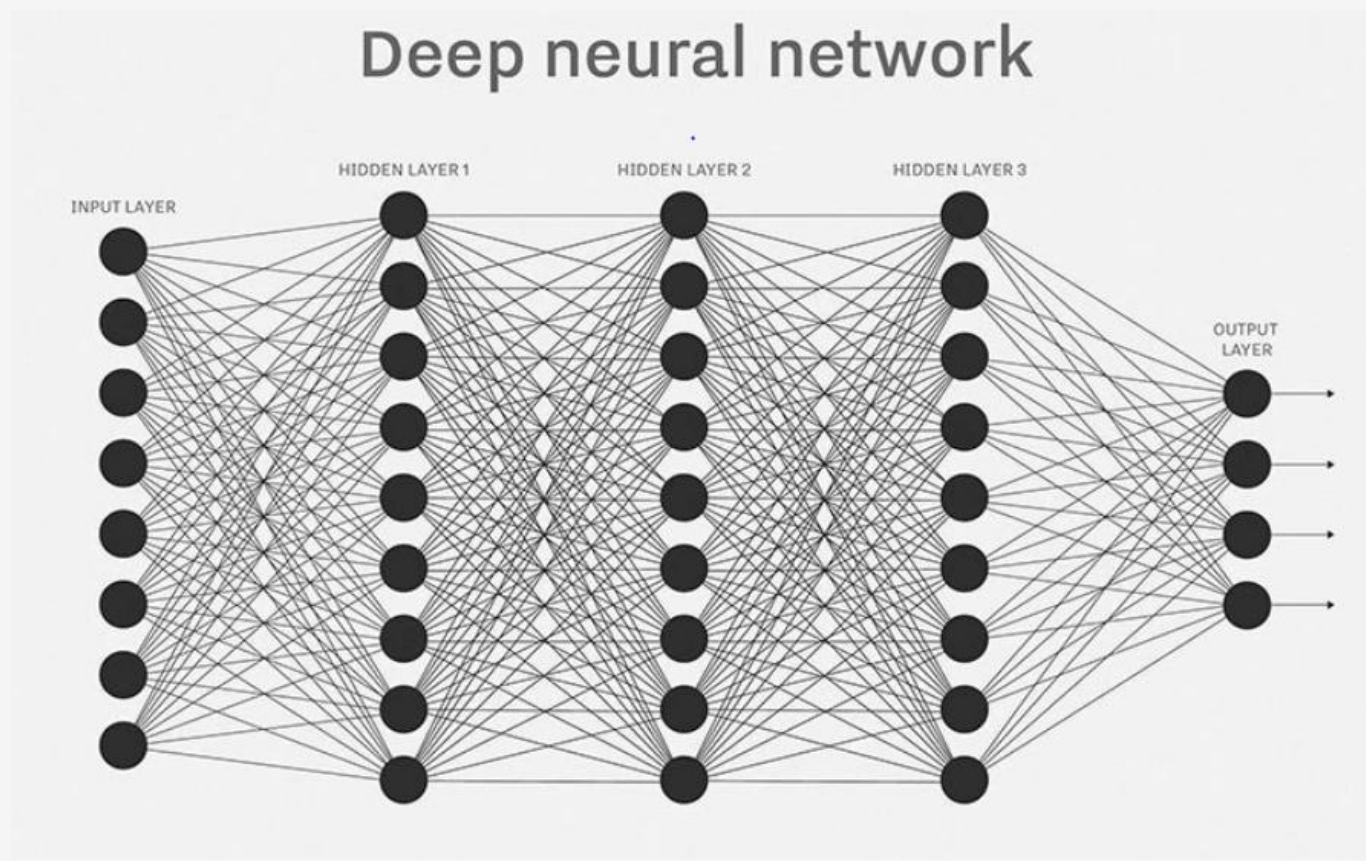
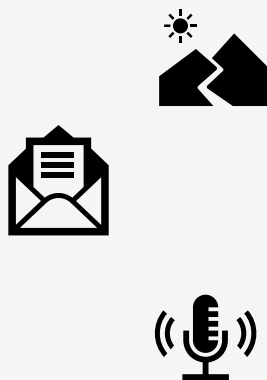
# Deep Learning Timeline



A-brief-history-of-ai-with-deep-learning

# AI가 똑똑한 이유?

AI가 똑똑한 이유는 엄청난 양의 데이터(이미지, 텍스트, 사운드, 동영상)를 초고속 컴퓨터(학습 능력)로 밤낮없이 읽으면서, 입력받은 모든 데이터의 패턴을 찾아냈기 때문(수능 만점자가 기출문제를 완벽하게 마스터해서 어떤 변형 문제가 나와도 맞히는 것과 비슷)



- Big Data
- Computing power
- Algorithm

# Supervised Learning(지도학습)



**Duck**



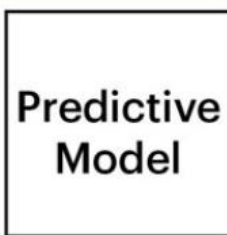
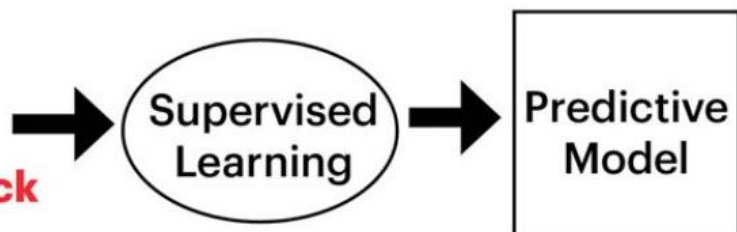
**Duck**



**Not Duck**



**Not Duck**



**Duck**



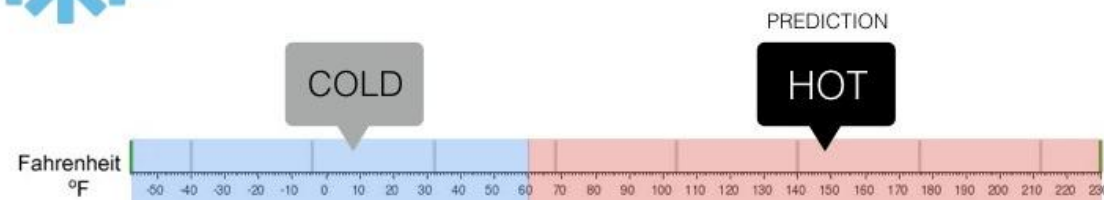
## Regression

What is the temperature going to be tomorrow?



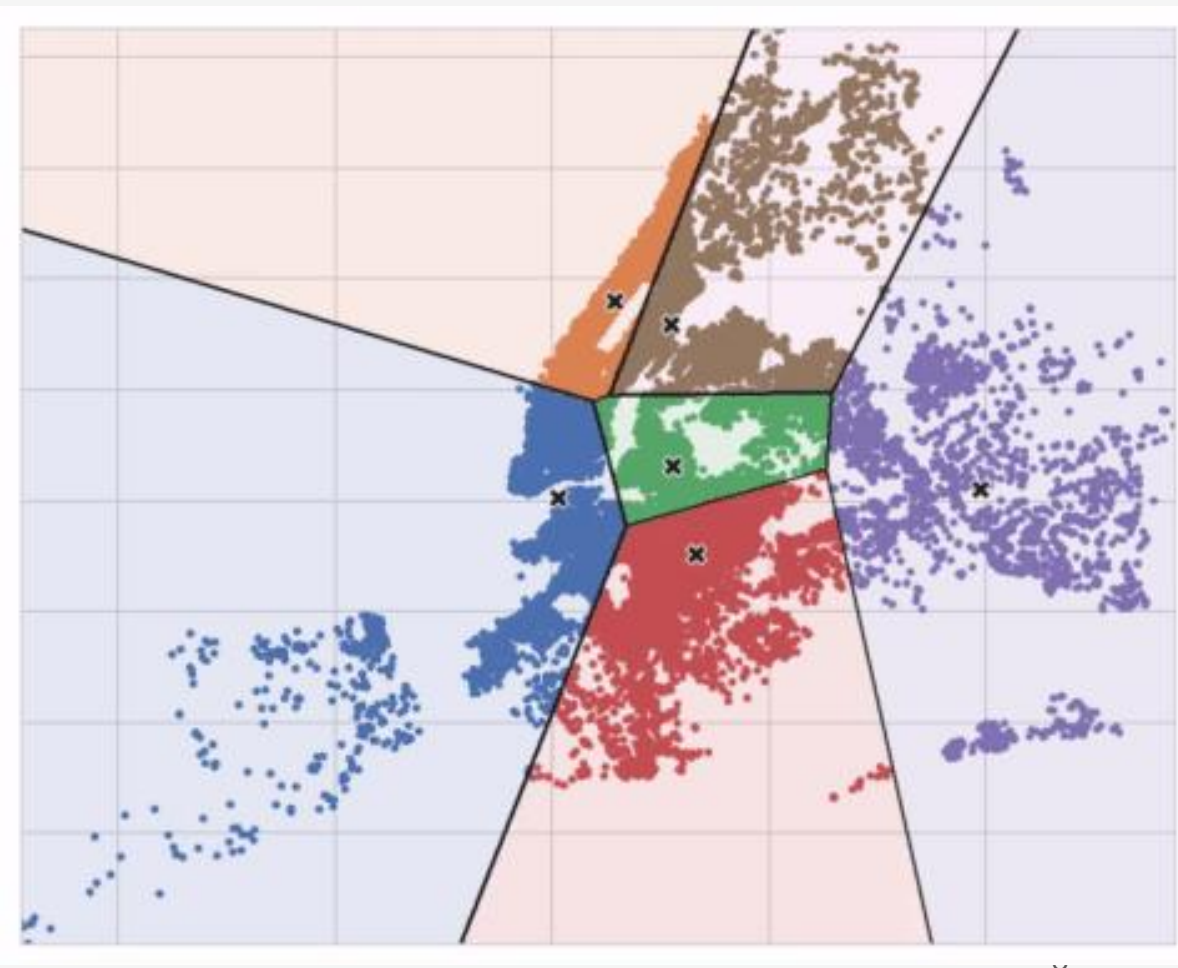
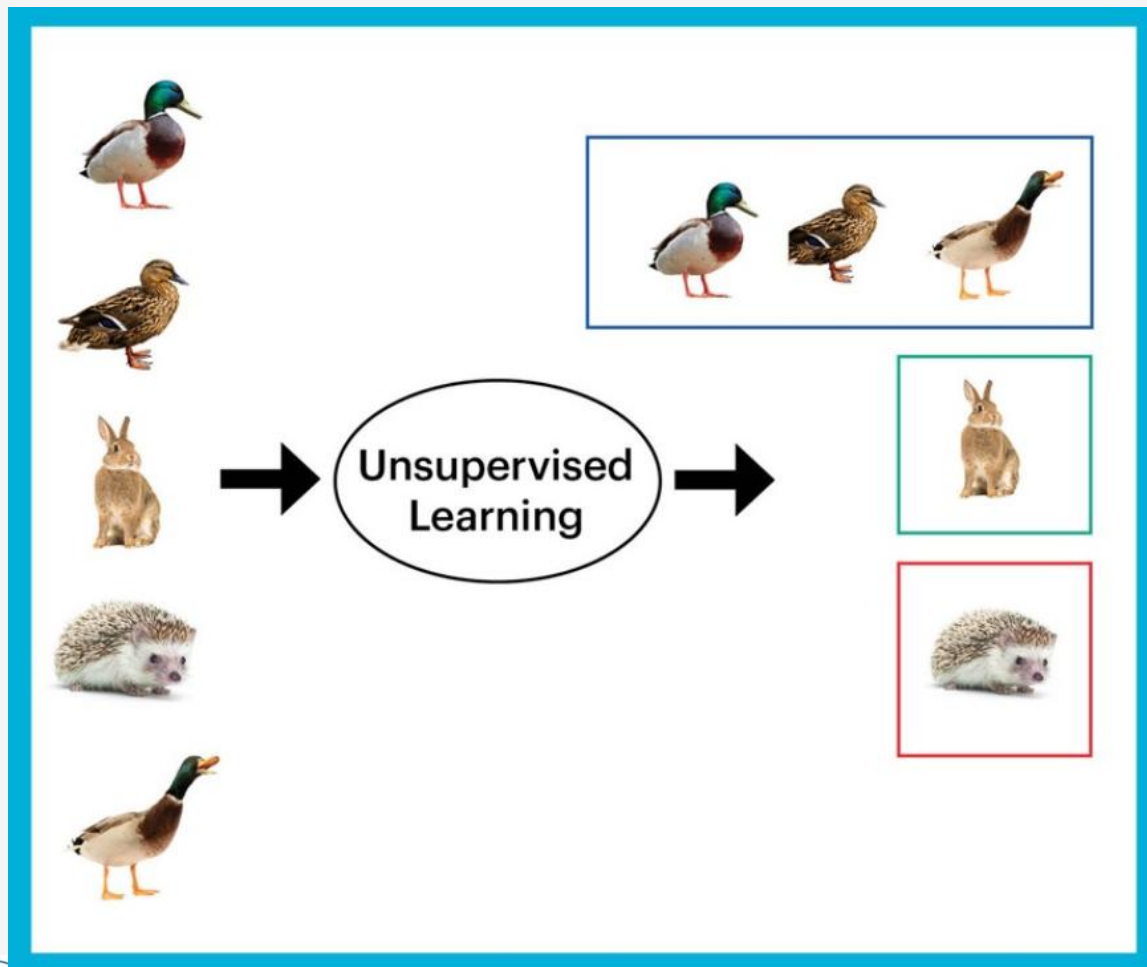
## Classification

Will it be Cold or Hot tomorrow?



# Unsupervised Learning(비지도학습)

## Cluster(군집)



# 목차

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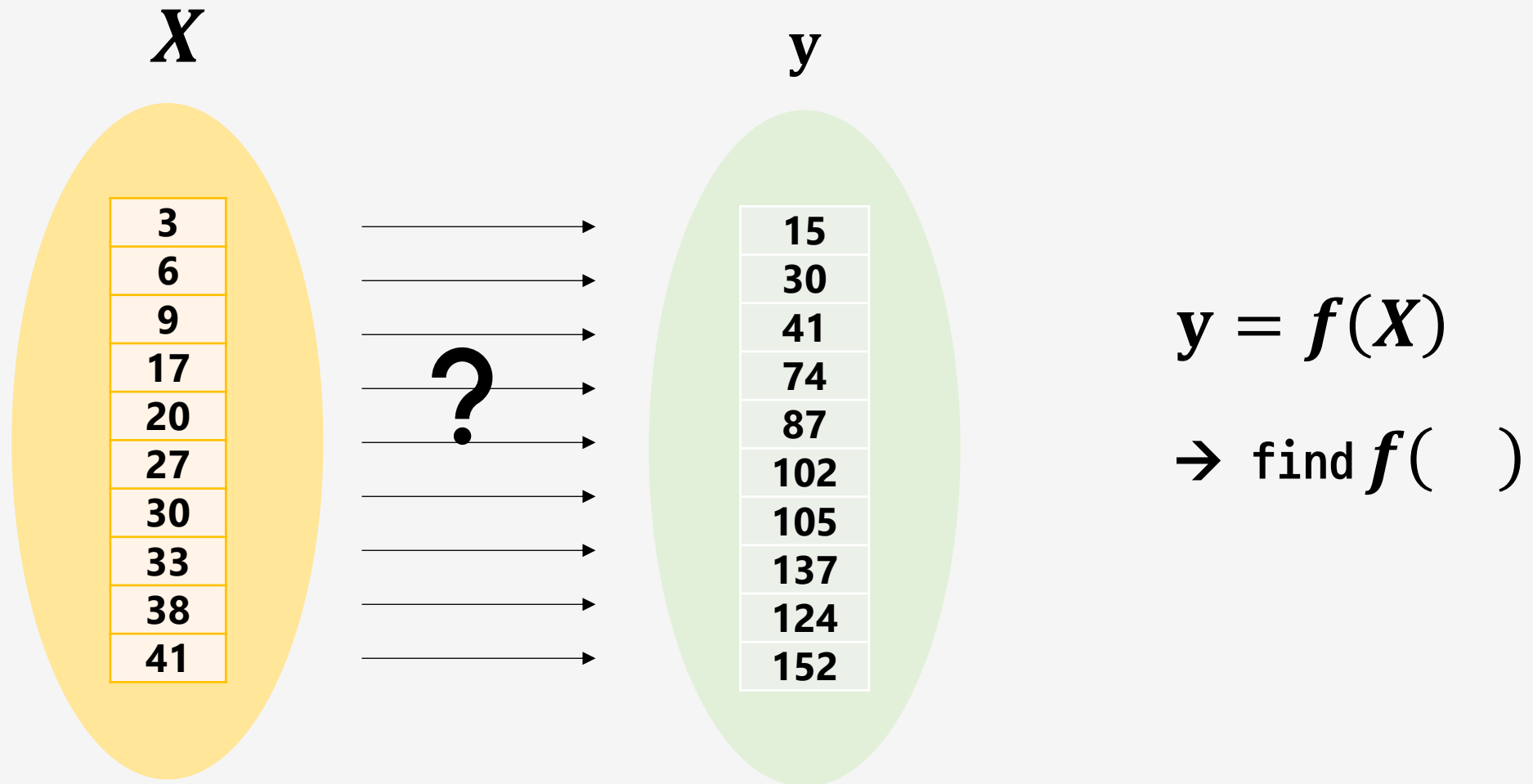
1. AI는 왜 똑똑하지?

2. AI는 어떻게 학습하지?

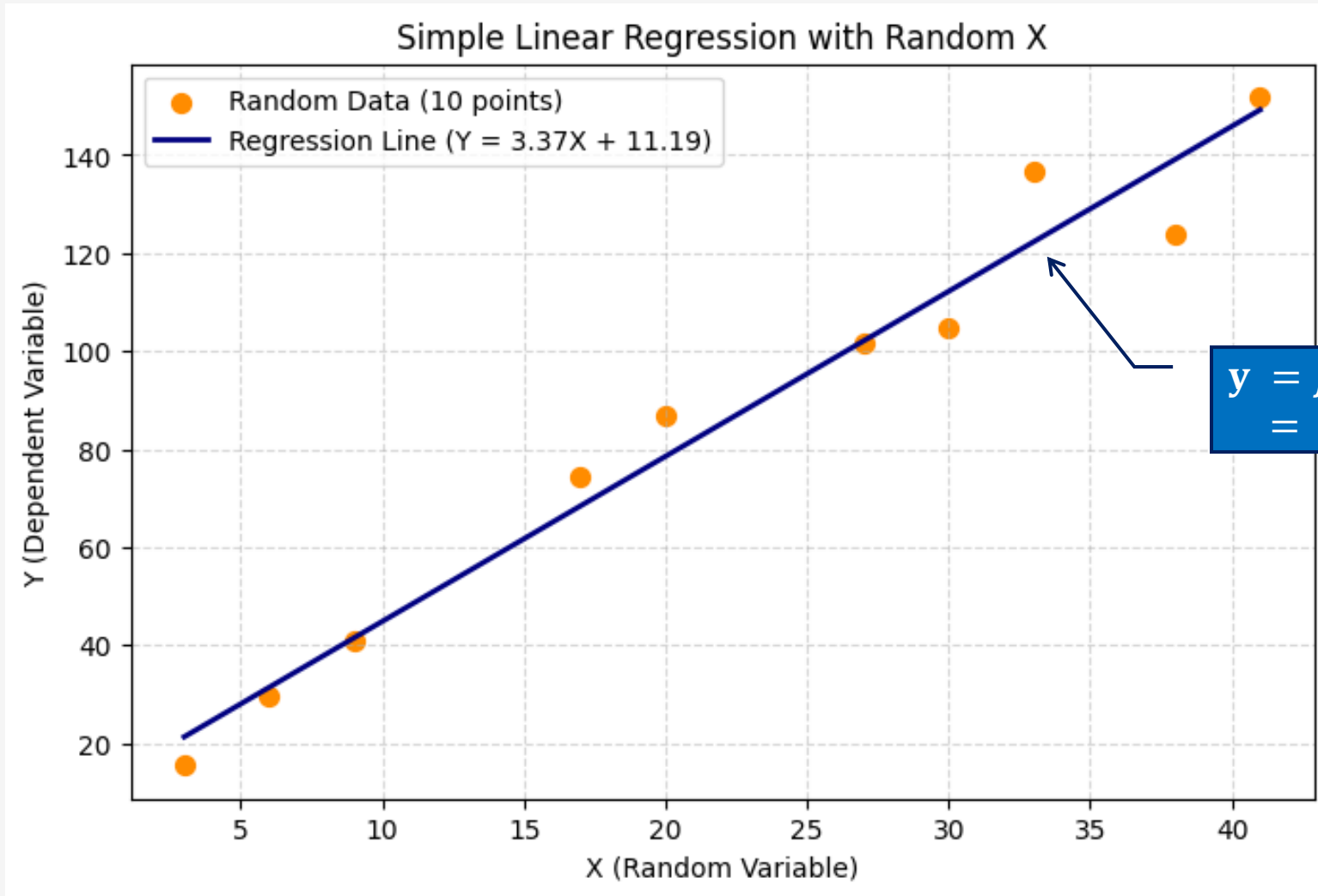
3. 생성형 AI

4. LLM agent

# AI는 어떻게 학습하지?



# AI는 어떻게 학습하지?



# AI는 어떻게 학습하지?

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$X$   $y$



wired.com

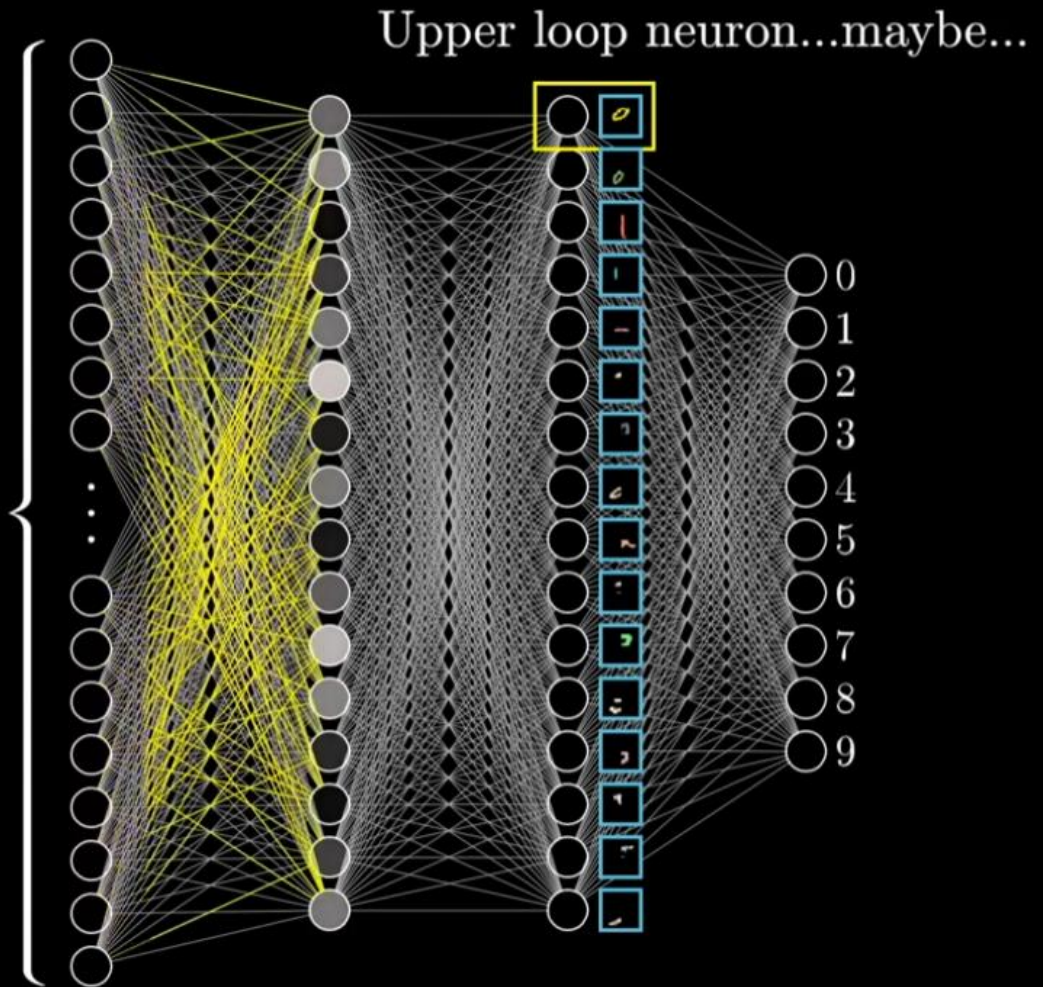
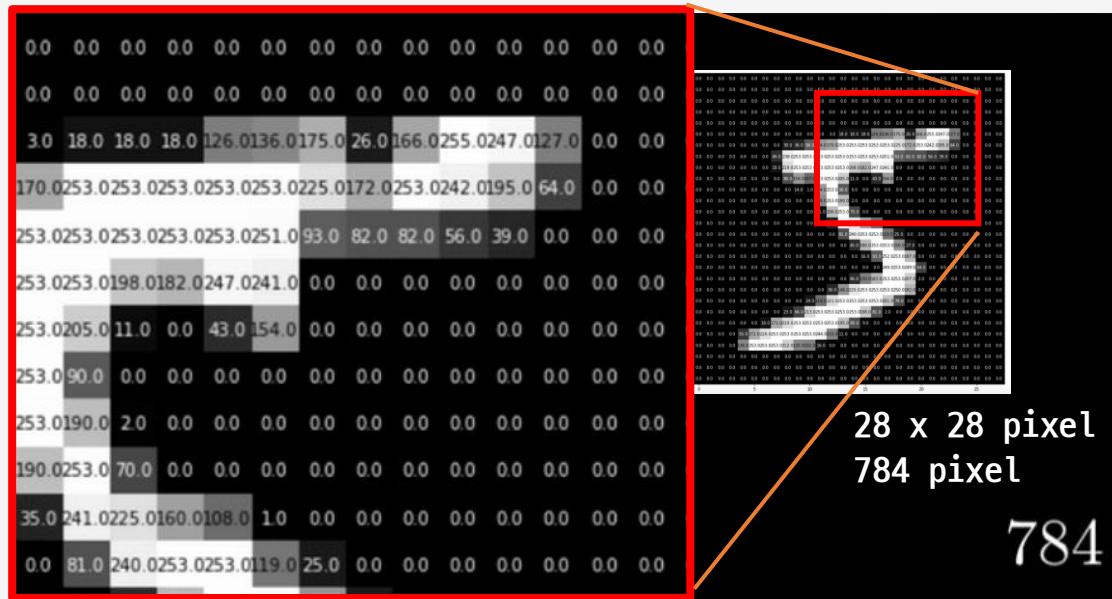
“Morpheus: The Matrix is everywhere. It is all around us. Even now, in this very room. You can see it when you look out your window or when you turn on your television. You can feel it when you go to work... when you go to church... when you pay your taxes. It is the world that has been pulled over your eyes to blind you from the truth.

Neo: What truth?

Morpheus: That you are a slave, Neo. Like everyone else you were born into bondage. Into a prison that you cannot taste or see or touch. A prison for your mind.”

— Lana Wachowski, [The Matrix: The Shooting Script](#)

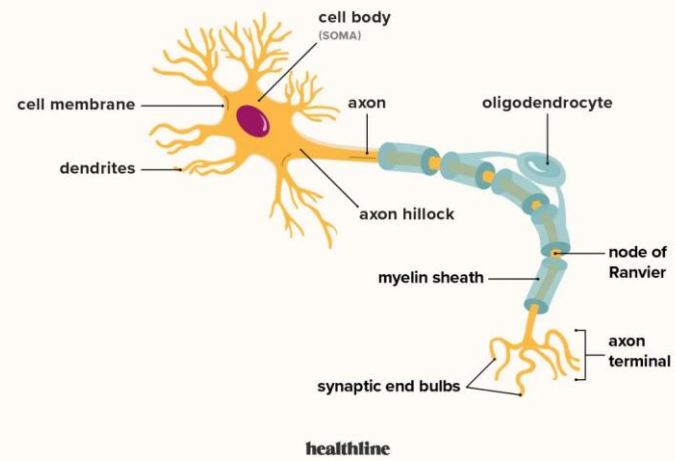
# What is a deep learning?



What is a neural network?(start= 3 min, end = 5 min)

# Human Neurons ?

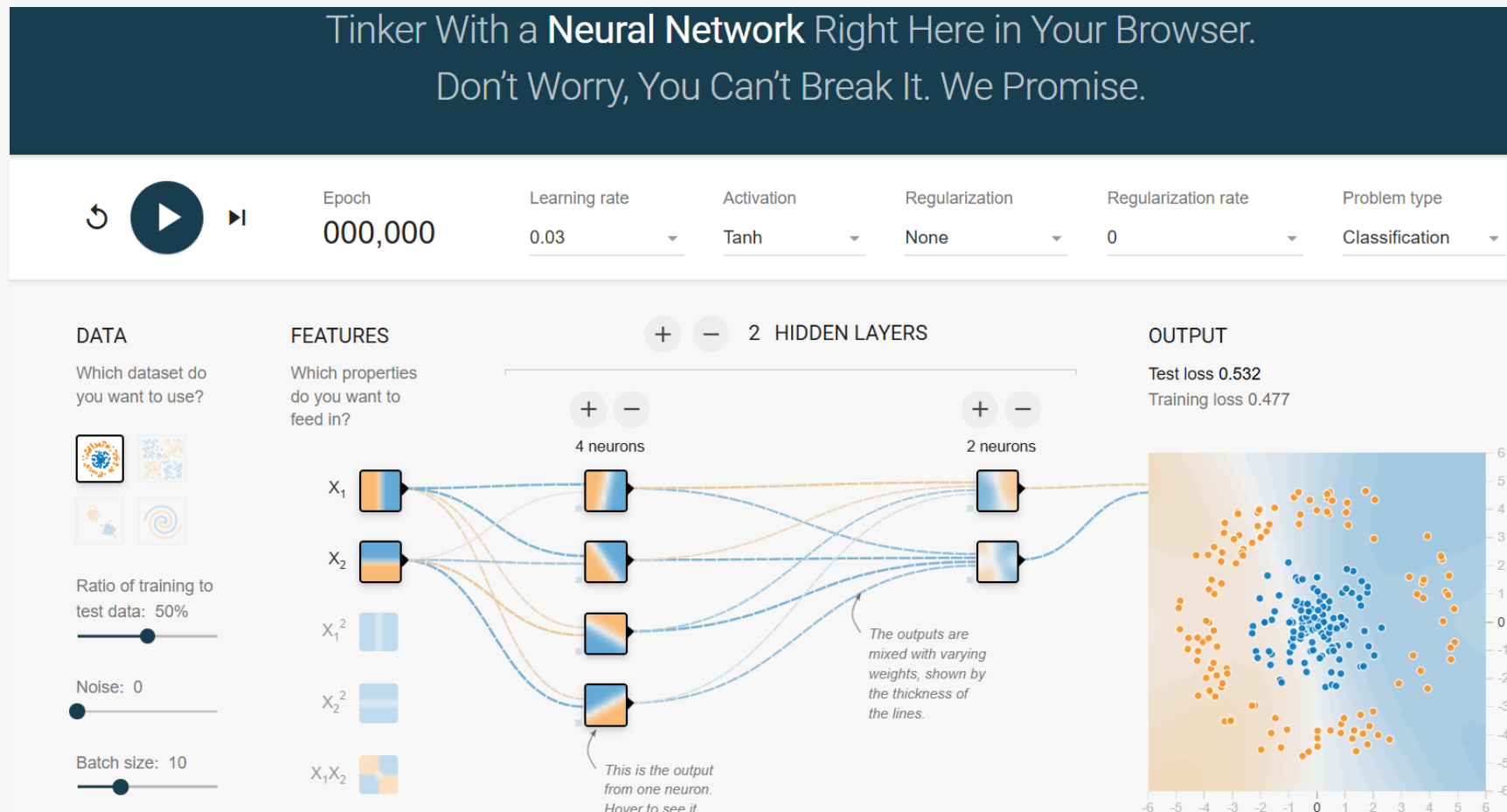
Structure of a neuron



<https://unsplash.com/>

# 실습하기

<https://playground.tensorflow.org/>



# 실습하기

## Word Embedding

|    | have   | the    | bards  | who    | preceded | me     | left   | any    | theme  | unsung |
|----|--------|--------|--------|--------|----------|--------|--------|--------|--------|--------|
| 0  | -1.621 | -0.634 | -0.324 | -1.357 | -0.261   | 4.632  | 1.236  | -0.046 | -1.518 | -0.082 |
| 1  | -1.401 | 0.683  | -0.114 | 1.891  | 0.544    | -1.487 | 1.247  | -0.408 | 0.202  | -0.022 |
|    | ...    | ...    | ...    | ...    | ...      | ...    | ...    | ...    | ...    | ...    |
| 49 | -0.647 | 0.200  | -0.203 | 1.573  | -0.520   | -0.355 | -1.491 | 1.325  | 2.550  | -0.010 |

<https://word2vec.kr/search/>



한국-서울+도쿄

QUERY

+한국/Noun

+도쿄/Noun

-서울/Noun

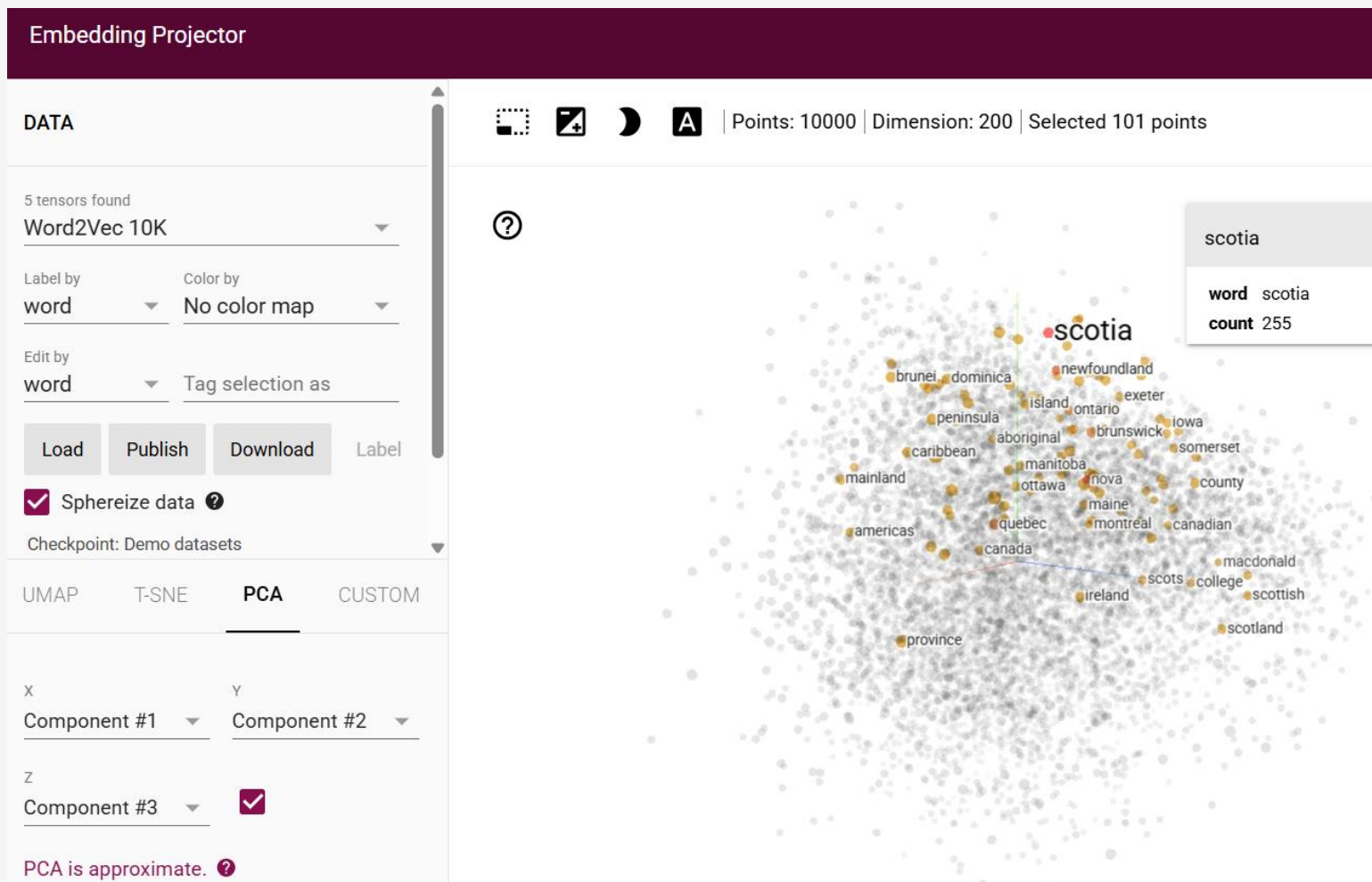
RESULT

일본/Noun

한국-프랑스+파리

# 실습하기

## tensorflow projector



# 목차

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# 대규모 언어 모델 (Large Language Model, LLM)

데이터를 학습한 딥러닝은 이전 데이터(단어, 픽셀 등)들을 기반으로 다음 데이터(단어, 픽셀)를 확률적으로 예측하여 텍스트, 이미지, 음악, 코드 등 새로운 콘텐츠를 생성(Generative)하는 AI

블랙

?

핑크

커피

홀



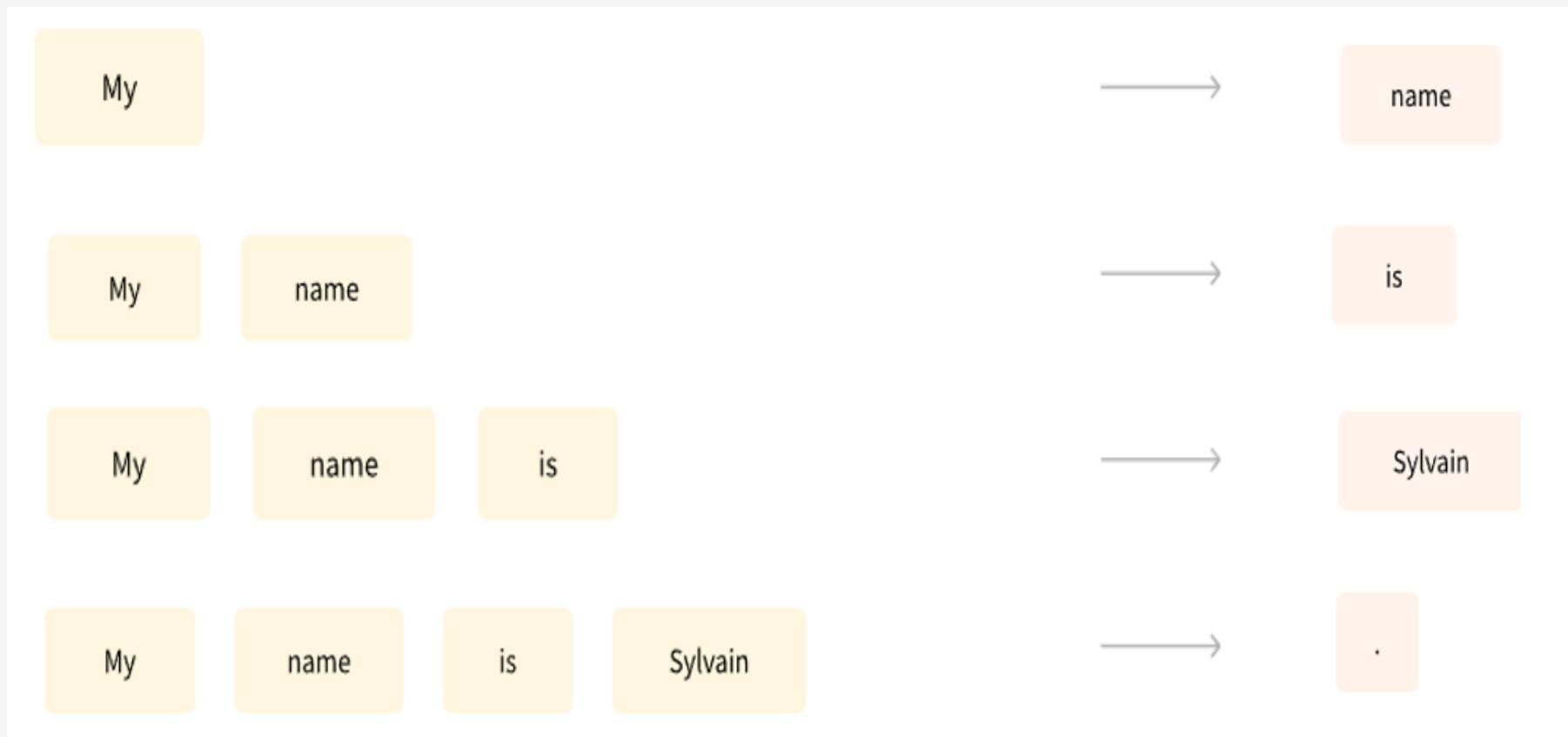
# 대규모 언어 모델 (Large Language Model, LLM)

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# 확률적 언어모델

데이터를 학습한 딥러닝은 이전 단어들을 기반으로 다음 단어를 확률적으로 예측하여 말을 이어가는 끝 말 이어가기 끝판 왕 → LLM(Large Language Model)



<https://huggingface.co/learn/nlp-course/chapter1/4?fw=pt>

My

# 확률적 언어모델을 가진 앵무새(Generative AI, LLM)

tokenizer



- 수십억 개의 단어를 학습하여 사람처럼 자연스럽게 언어를 생성하는 인공지능
- 신경망(Neural Network) 기반으로 문맥을 이해하고 논리적 답변 생성 가능
- 대표 모델: GPT-4, Claude, Gemini, LLaMA 등

# LLM과 인간과 같은 감정으로 소통하진 마라

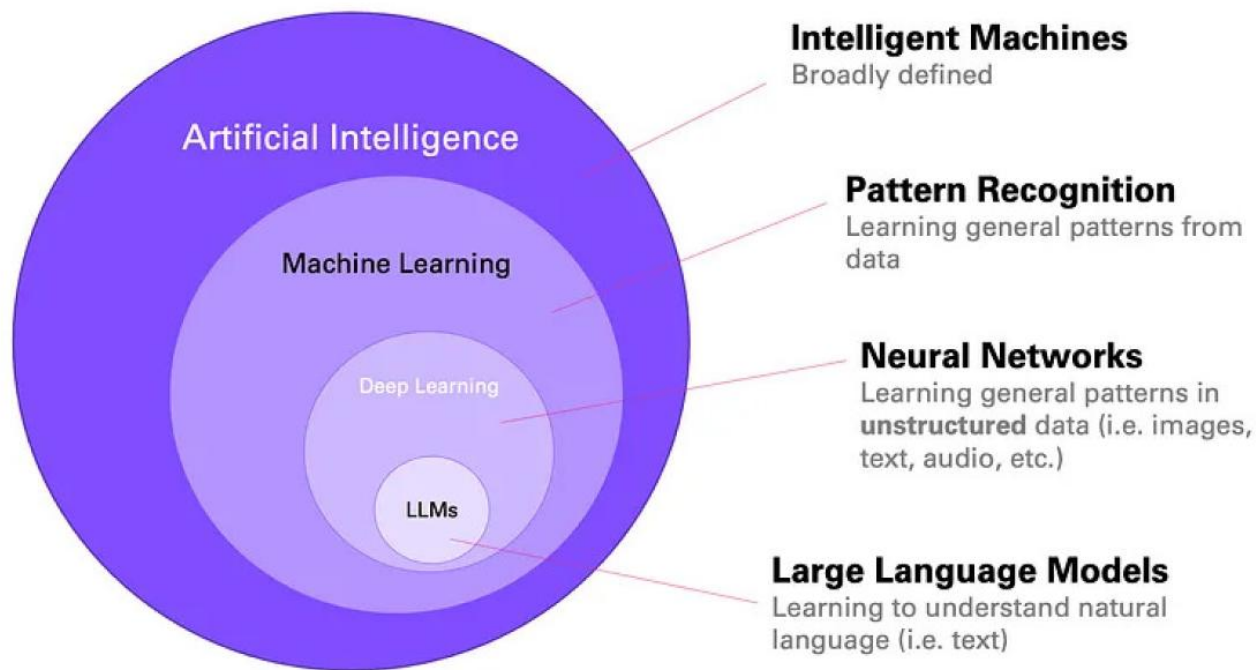
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- ❖ Gary Marcus: "LLMs are fluent bullshitters."
  - ❖ 언어적 유창성은 뛰어나나 사실 검증 메커니즘이 없음
- ❖ Yejin Choi: "Commonsense missing, therefore bullshit-prone"
  - ❖ 세상에 대한 상식 없이 말하기 때문에 자주 현실과 동떨어짐
- ❖ 코마에서 깨어난 신입사원 아인슈타인

AI를 두려워하지 말고  
도구로써 잘 이용하자

# 대규모 언어 모델 (Large Language Model, LLM)

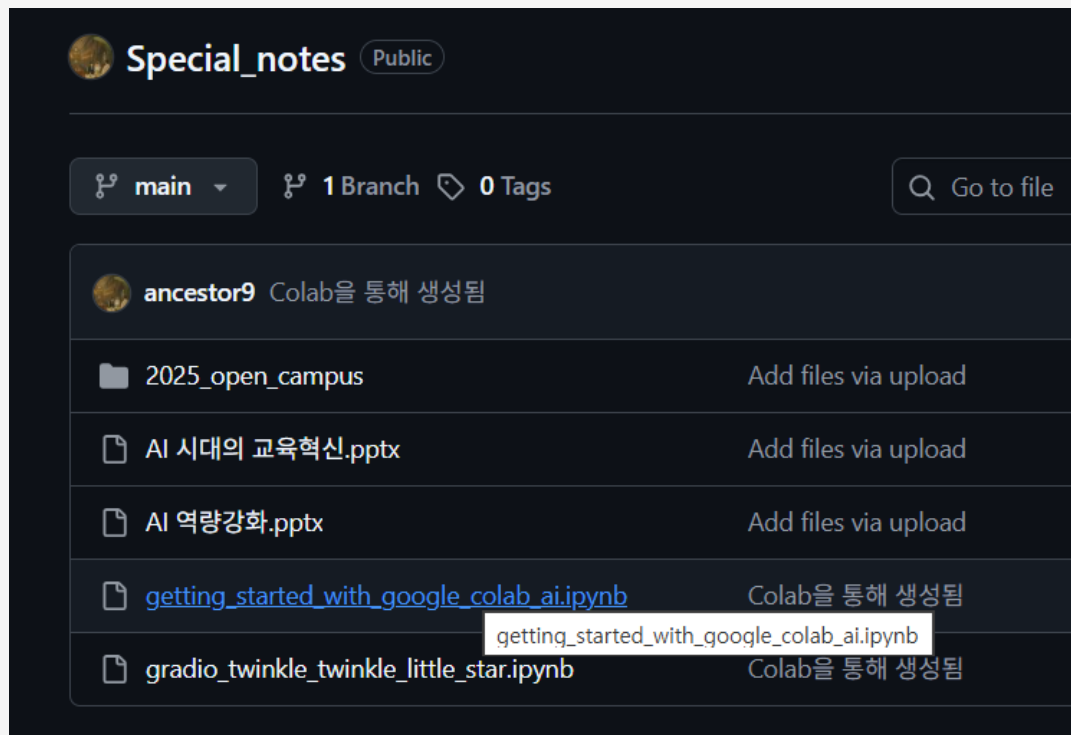
- ❖ 수십억 개의 단어를 학습하여 사람처럼 자연스럽게 언어를 생성하는 인공지능
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- ❖ 대표 모델: GPT-4, Claude, Gemini, LLaMA 등



# 실습하기

## 제미나이 AI 실습 코드(colab)

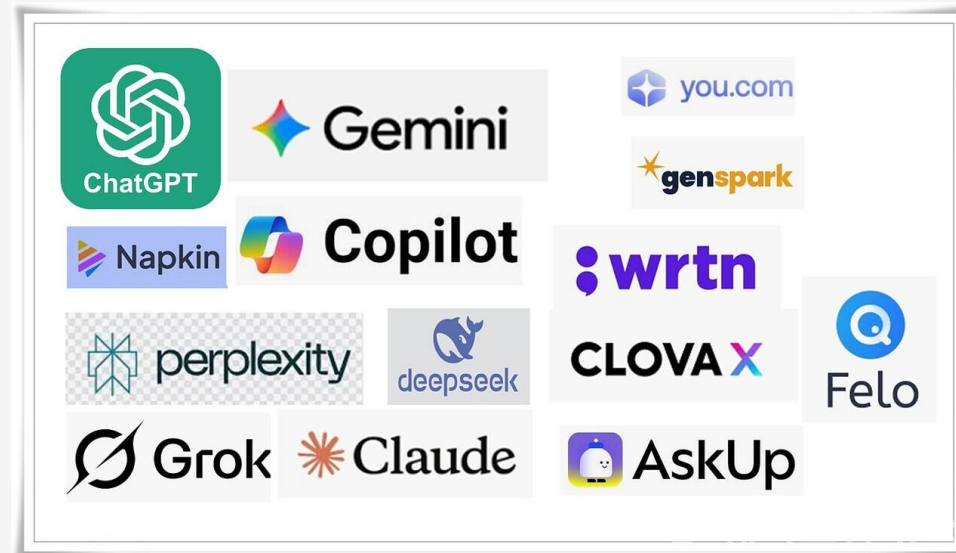
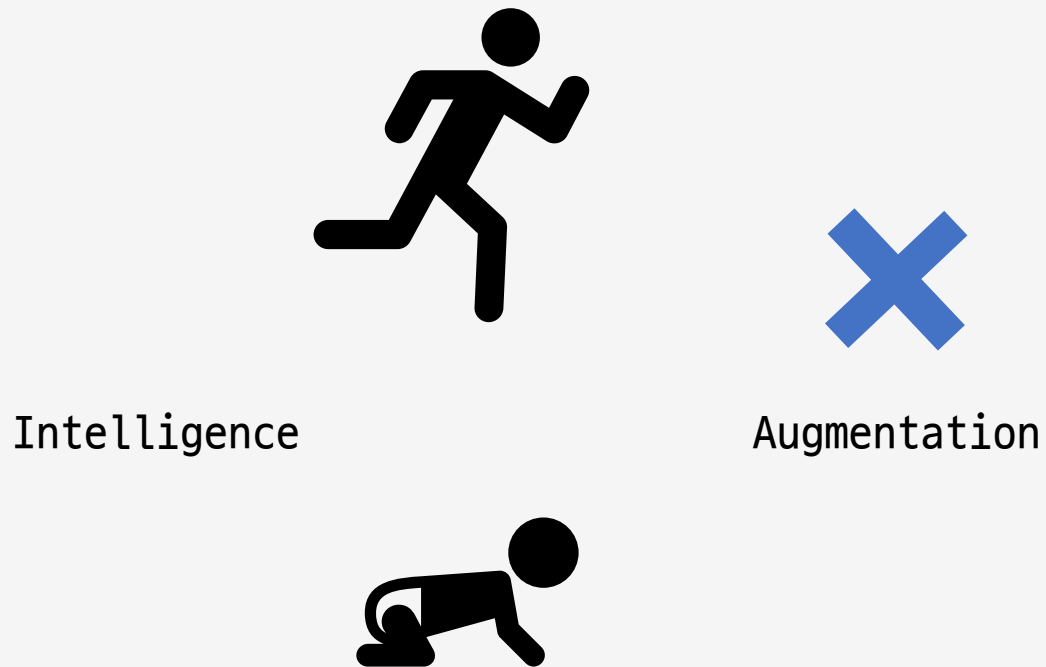
1. 텍스트 생성(Generate text)
2. 번역하기(Translate languages)
3. 소설 만들기(Write creative content)
4. 보이스피싱 탐지(Categorize text)



# To be or not to be

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AI(Artificial Intelligence) < IA (Intelligence Augmentation)



# 목차

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# 실습하기

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## 제미나이 AI 실습하는 코드(colab python)

- 1. 텍스트 생성(Generate text)
- 2. 번역하기(Translate languages)
- 3. 소설 만들기(Write creative content)
- 4. 보이스피싱 탐지(Categorize text)



4가지 서비스를 친구에게 휴대폰이나  
인터넷으로 제시하자 ! using [gradio.app](https://gradio.app)



gradio를 사용하여 4가지 기능을 만들어줘

# LLM Agent

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로봇청소기는 Agent이고, 자율주행차는 AI Agent이며, 사람처럼 대화하며 인터넷 검색을 하고 엑셀을 편집해 보고서를 완성해 주는 AI는 LLM Agent (Agent  $\supset$  AI Agent  $\supset$  LLM Agent)

LLM agent is a large language model embedded in a loop of reasoning, acting, and feedback, where it can call external tools and adapt its behavior based on results.

# LLM

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언어 모델은 단어 시퀀스에 확률을 할당(assign) 하는 일을 하는 모델로 가장 자연스러운 단어 시퀀스를 찾아내어 예측하는 모델



LLMs → *Brain*  
(plain language model)

- ❖ It talks.
  - you send it a message, it send text back.
- ❖ It just generate text and predict text. (It can't check your email or it can't search the Web...)

# AI Agent

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AI Agent is simply just a language model that can use tools that's running in a loop until it finishes a job.



Agent

→ Hands

(GIVE THIS MODEL 2 NEW ABILITIES)

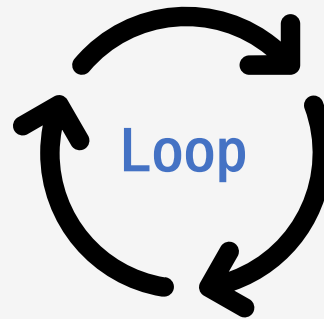
- ❖ 1st Ability
  - take actions in the real world (Tool-calling)
- ❖ 2nd Ability
  - to keep going on its own step by step, instead of stopping after just a single reply

# AI Agent

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**1 THINK**  
SYSTEM PROMPT & GOAL

**4 THINK AGAIN**  
THE MODEL LOOKS AT THE RESULT  
AND DECIDES WHAT TO DO NEXT



**2 ACT**  
MODEL DECIDES TO USE A TOOL  
AND SENDS THE ACTION

**3 OBSERVE**  
CODE RUNS THE TOOL AND RETURN THE  
RESULT. THE MODEL OBSERVE THE  
RESULT

# Tool-Calling

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## TOOL-CALLING (The Hands)

- ❖ It generates text that tells whatever software it's working inside of to call a tool.

# Tool-Calling

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## Context Window (Short-term working Memory)

- ❖ The maximum amount of text it can process and remember at any single time during a conversation.
- ❖ Conversation history is tracked up to LLM' Context Window.

# Prompt, Message

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## System Prompt

- ❖ Is where you set the rules. So you tell the agent who it is, how to behave.  
General rules things that it should do.

## User Messages

- ❖ The input from the person or the system.

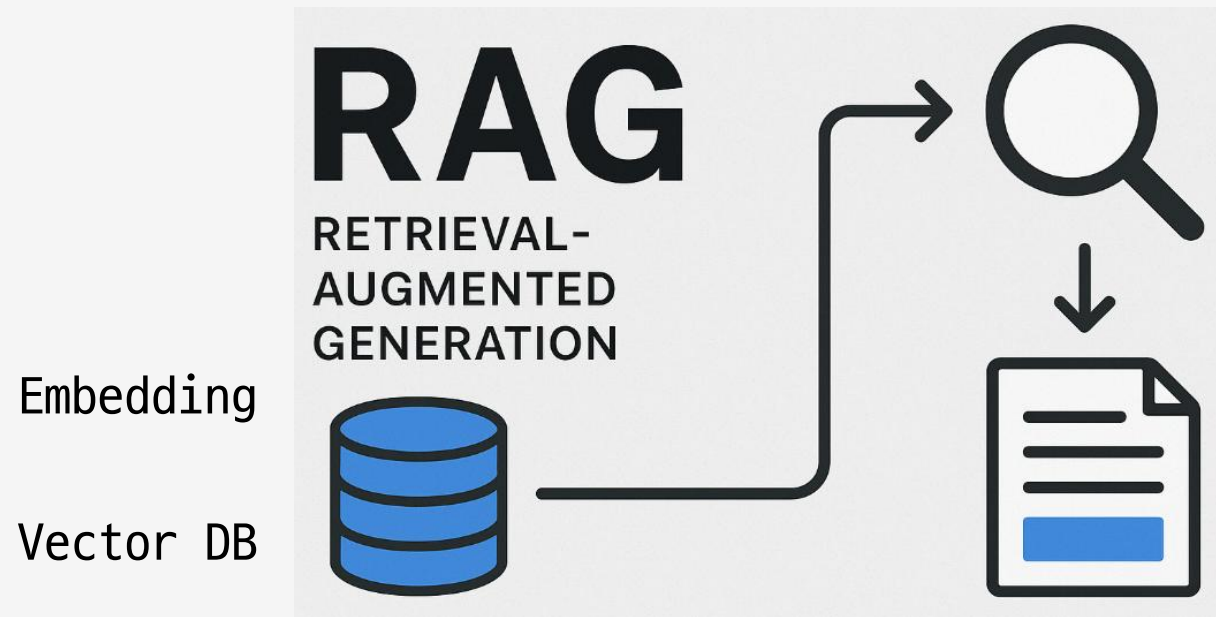
## Assistant Messages

- ❖ What the model says back.

# RAG

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Conversation history can get quite large and it can actually **not fit** inside of the current **context window** → RAG



# How to build AI Agent

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## 1 NO-CODE

Build an agent by filling out forms or dragging blocks around.

## 2 LOW-CODE

Write parts of the agent like a visual canvas.

## 3 AGENT HARNESESSES

Install a complete powerful agent and customize it.

## 4 FULL CODE

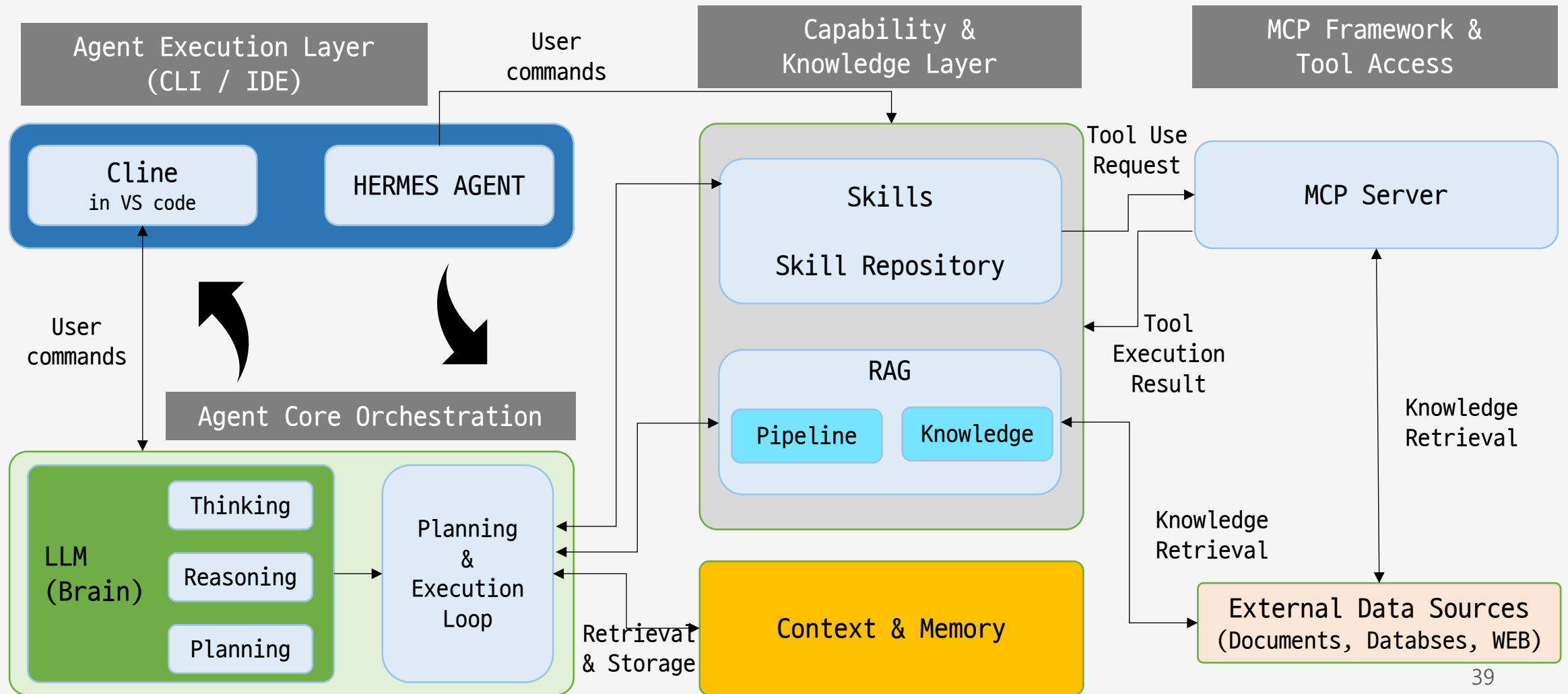
Writing the entire thing in a programming language like Python.

# How to build AI Agent

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| Level |                  | Tools                                                                                |
|-------|------------------|--------------------------------------------------------------------------------------|
| 1     | NO-CODE          | ▪ Genspark, Lindi, Gumloop, Stack AI, Relevance AI, Botpress, Voiceflow              |
| 2     | LOW-CODE         | ▪ n8n, make, KNIME, Flowise, Langflow, Dify, Activepieces                            |
| 3     | AGENT HARNESSSES | ▪ Cline, Antigravity, Claude CLI, Codex (Coding Agents)<br>▪ OpenClaw, HERMES AGENTS |
| 4     | FULL CODE        | ▪ Python with Langchain, Langgraph                                                   |

# LLM Agent Architecture



# Practice with google Agent SDK

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